

analysis about systematic withdrawals will fundamentally tie into a few basic formulas. First, using the notation of Microsoft Excel, the PMT formula allows us to calculate the sustainable amount of spending in retirement. This formula is defined as:

=PMT(rate,nper,pv,fv,type)

The variable **rate** is the assumed investment return in decimal form, **nper** is the number of periods (I usually think in terms of years) that the nest egg should last, **pv** is the amount you have saved at retirement, and **fv** is the amount you wish to be able to take out for a legacy at the end of retirement (counterintuitively, it needs to have a negative sign if you want to have money left over at the end). It is more conservative to set **type** to a value of 1 to reflect that you will take your distributions at the start of each time period. For instance,

=PMT(0.0133,30,1000000,0,1)

will provide you with an answer of \$40,109, meaning if you assume an inflation-adjusted compounding investment return of 1.33 percent, want your money to last thirty years, start with \$1 million, want to end up with \$0, and take your distributions at the start of each year, then you can sustainably spend \$40,109, with subsequent inflation adjustments for your yearly spending amount. I say that inflation adjustments are allowed because I am thinking of the 1.33 percent return as a real or after-inflation return. As a percentage of initial assets, that is just slightly above a 4 percent spending rate.

With all else being equal, if you increase the value of **rate** (your investment return), then you can increase the sustainable spending amount. For instance, replace 1.33 percent with 7 percent in the function, and the spending answer jumps to \$75,314. This represents a 7.53 percent withdrawal percentage from initial assets. Perhaps surprisingly as well, with a 7 percent return assumption, if you wanted to fully preserve the purchasing power of your initial retirement portfolio after thirty years (which requires putting in a value of -1000000 for **fv**), the sustainable spending rate falls only to 6.54 percent.